		(0.45 ± 0.02) kg
2	**C**	The general equation is $F = kx \rightarrow x = \frac{F}{k}$ Let W represent the weight of each mass. For the upper spring, $x_{upper} = \frac{2W}{30}$ For the lower spring, $x_{lower} = \frac{W}{10}$ $\frac{x_{upper}}{x_{lower}} = \frac{2}{3}$ $\frac{2.0}{x_{lower}} = \frac{2}{3} \rightarrow x_{lower} = 3.0$ cm
3	**C**	$F_R = \frac{d(mv)}{} = v\frac{dm}{} = (20)(0.50) = 10$ N